

2023 AMC 10A Problem 25 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 25

Problem:

If A and B are vertices of a polyhedron, define the distance $d(A, B)$ to be the minimum number of edges of the polyhedron one must traverse in order to connect A and B . For example, if $\overline{AB}$ is an edge of the polyhedron, then $d(A, B) = 1$, but if $\overline{AC}$ and $\overline{CB}$ are edges and $\overline{AB}$ is not an edge, then $d(A, B) = 2$. Let Q , R , and S be randomly chosen distinct vertices of a regular icosahedron (regular polyhedron made up of 20 equilateral triangles). What is the probability that $d(Q, R) > d(R, S)$?

Answer Choices:

A. $\frac{7}{22}$

B. $\frac{1}{3}$

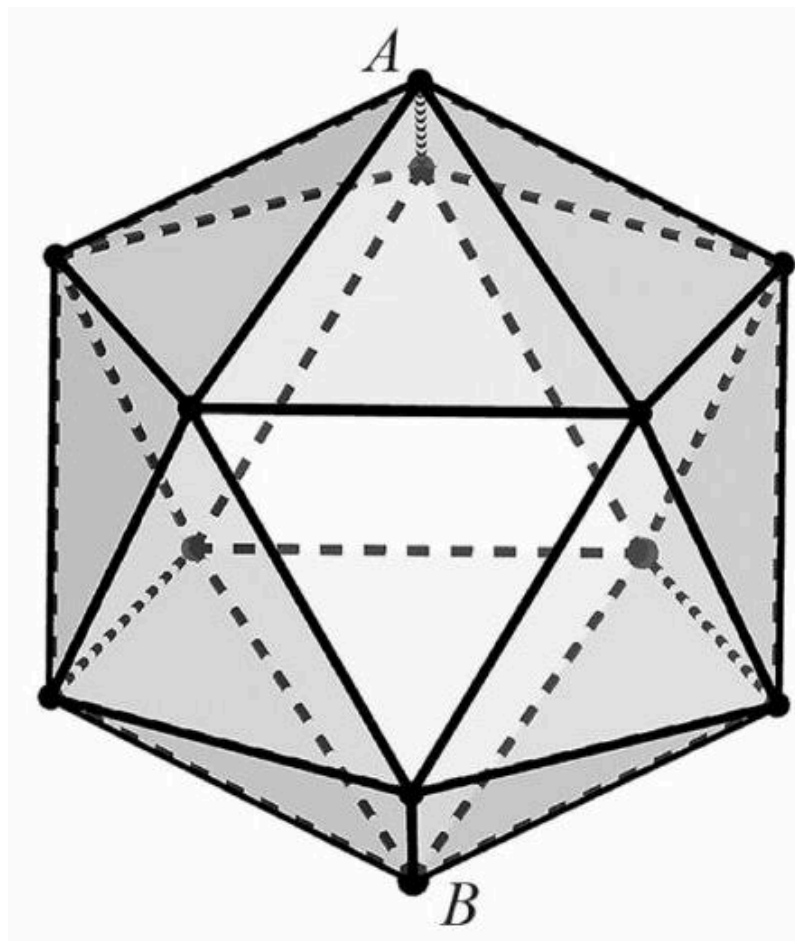
C. $\frac{3}{8}$

D. $\frac{5}{12}$

E. $\frac{1}{2}$

Solution:

One way to envision a regular icosahedron is with pentagonal pyramids on top and bottom, joined by a pentagonal antiprism. Let vertex A be at the top, as illustrated.



There are five vertices on the base of the pyramid with apex A , so they are all at distance 1 from A . There are five vertices on the base of the pyramid with apex B , which are all at distance 1 from the previous five vertices, so they are at distance 2 from A . Finally, only vertex B is at distance 3 from A . Thus from any vertex, there are 5 vertices at distance 1, 5 at distance 2, and 1 at distance 3. Randomly select Q , R , and S from the 12 vertices. Let $\mathbb{P}[E]$ denote the probability of event E , and let $\mathbb{P}[E \mid F]$ denote the probability of event E given event F . By symmetry

$$\mathbb{P}[d(Q, R) > d(R, S)] = \mathbb{P}[d(Q, R) < d(R, S)]$$

so it suffices to compute $\mathbb{P}[d(Q, R) = d(R, S)]$ and divide its complement by 2 to find $\mathbb{P}[d(Q, R) > d(R, S)]$. Note that $\mathbb{P}[d(Q, R) = 1] = \frac{5}{11}$ and

$$\mathbb{P}[d(R, S) = 1 \mid d(Q, R) = 1] = \frac{4}{10} = \frac{2}{5}$$

because vertex Q is not available. Therefore

$$\mathbb{P}[d(Q, R) = d(R, S) = 1] = \frac{5}{11} \cdot \frac{2}{5} = \frac{2}{11}$$

Because the number of vertices that are at distance 2 from Q is also 5, $\mathbb{P}[d(Q, R) = d(R, S) = 2]$ is also $\frac{2}{11}$. Note that $d(Q, R) = d(R, S) = 3$ is impossible, because Q and S are distinct. Therefore

$$\mathbb{P}[d(Q, R) > d(R, S)] = \frac{1}{2} \left(1 - \frac{2}{11} - \frac{2}{11} \right) = (\text{A}) \boxed{\frac{7}{22}}$$

OR

Use the same setting and notation as in the first solution. If $d(Q, R) = 1$, then $d(Q, R) \not> d(R, S)$. If $d(Q, R) = 3$, then $d(Q, R) > d(R, S)$, and this happens with probability $\frac{1}{11}$. If $d(Q, R) = 2$, then $d(Q, R) > d(R, S)$ exactly when $d(R, S) = 1$, which happens for 5 vertices of the remaining 10 from which to choose. Because $\mathbb{P}[d(Q, R) = 2] = \frac{5}{11}$,

$$\mathbb{P}[d(R, S) = 1 \mid d(Q, R) = 2] = \frac{5}{10} = \frac{1}{2}$$

so

$$\mathbb{P}[d(Q, R) > d(R, S) \text{ and } d(Q, R) = 2] = \frac{5}{11} \cdot \frac{1}{2} = \frac{5}{22}$$

The two ways found are mutually exclusive, so $\mathbb{P}[d(Q, R) > d(R, S)] = \frac{1}{11} + \frac{5}{22} = (\text{A}) \boxed{\frac{7}{22}}$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)